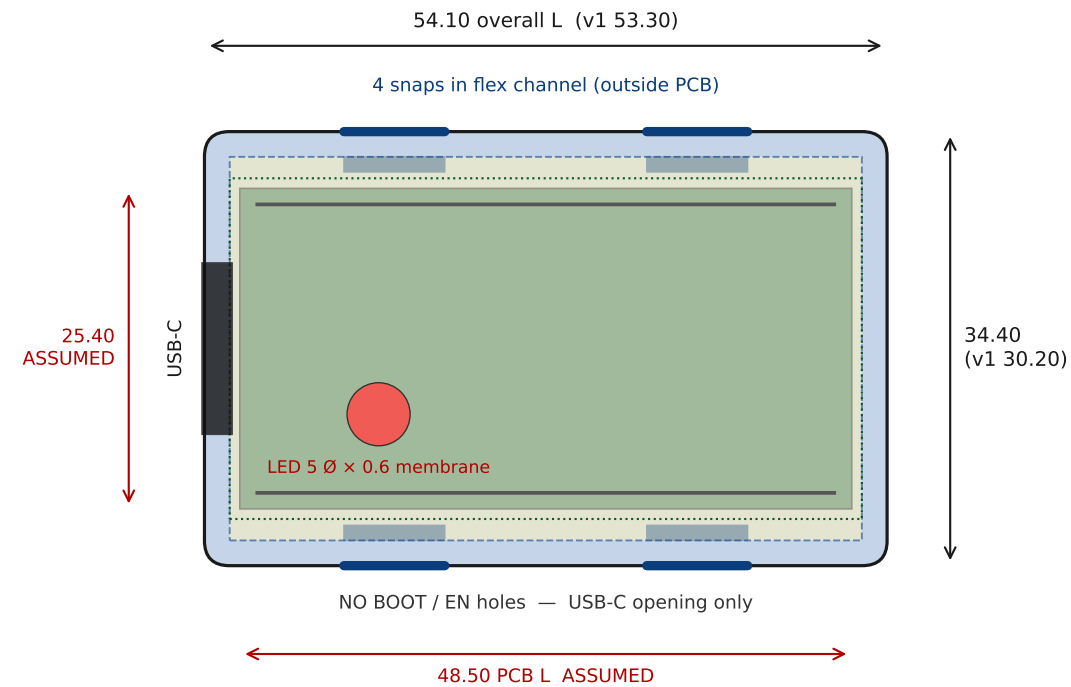
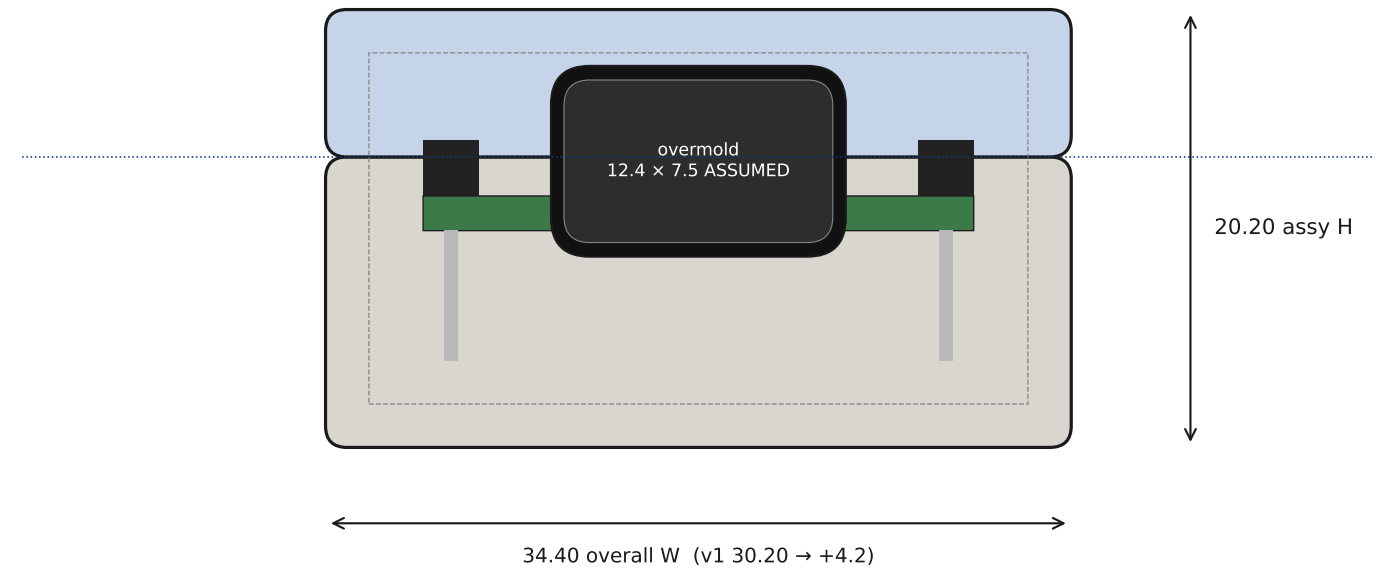


ESP-32 smoker relay snap case v2 — measured drawing (mm) — SNAP FIT FIX

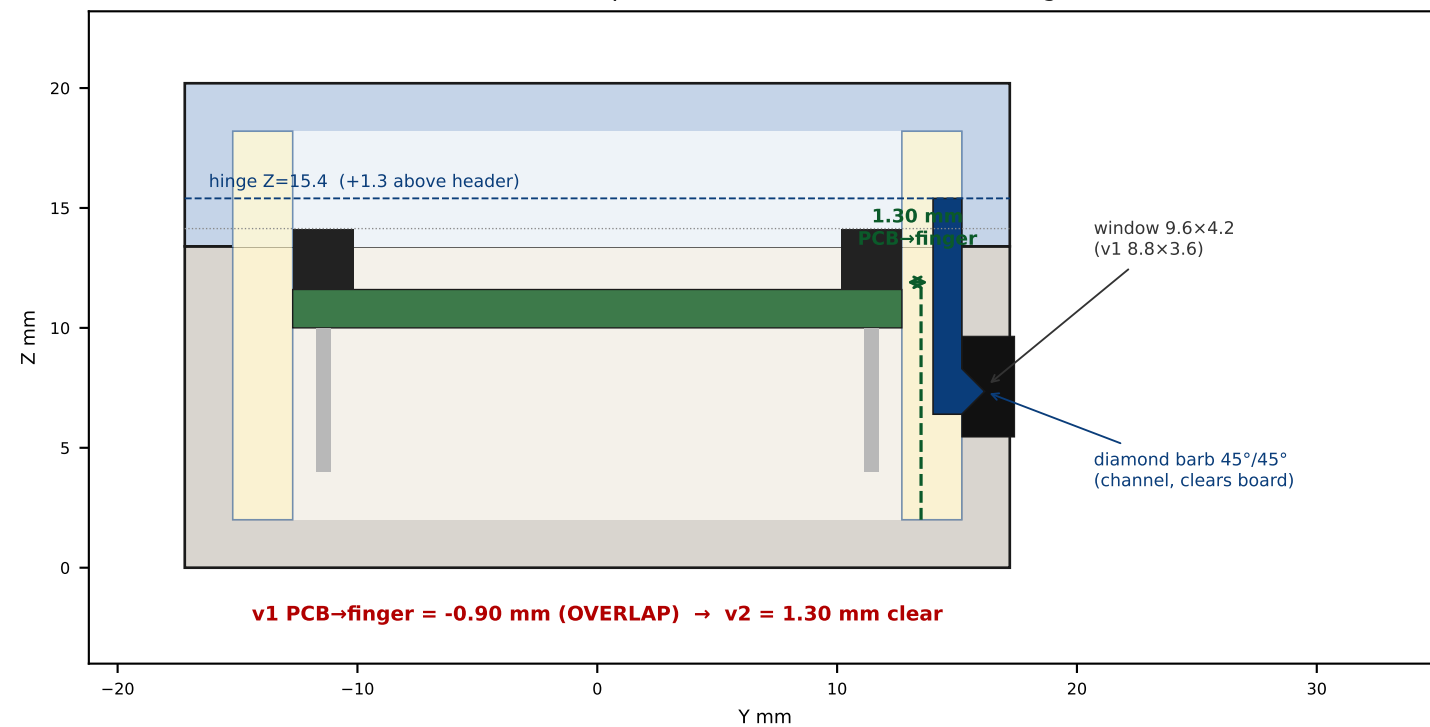
TOP (lid on, USB -X) — snap channel outside PCB



FRONT USB end (−X) opening 13.6×8.8



SECTION A-A snap channel (PCB clearance + hinge raise)



v2 fit numbers & delta vs v1

Item	v2	vs v1
PCB → snap finger inner	1.30 mm	v1 -0.90 (overlap)
Snap hinge vs header top	+1.26 mm	v1 hinge at rim (below header)
Barb catch vs board top	3.30 mm below	engages after stem clears board
Barb tip Z / catch Z	6.40 / 8.30	HOOK_L 7.0 (v1 6.2)
XY pocket clear	0.80 mm	v1 0.40 (+0.40)
Inner W / Outer W	30.40 / 34.40	v1 26.20 / 30.20
Base windows	9.6 × 4.2	v1 8.8 × 3.6
Base changed?	YES	wider cavity + corner stops + bigger windows
USB / BOOT / EN	USB-C only	no BOOT/EN holes; LED membrane kept
Snap / screws	4 diamond / 0	K1C, PETG, 45°/45°, no supports

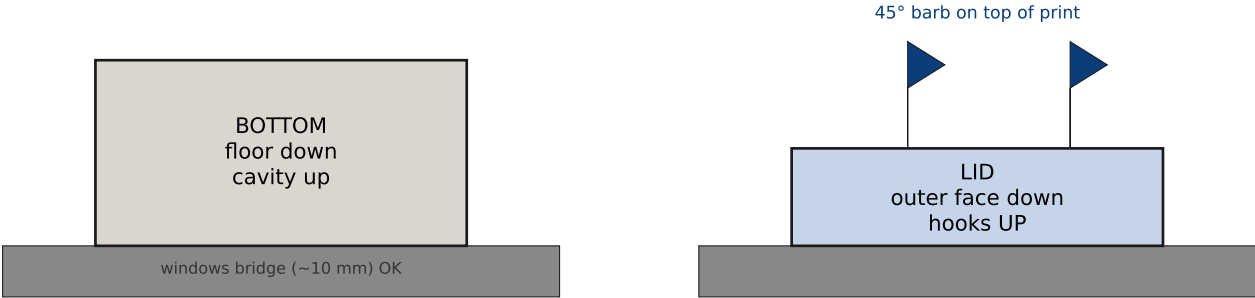
ASSUMED board envelope unchanged from v1. Same 3 caliper flags. Fit fix = snap channel + raised hinge + wider windows + +0.4 mm pocket.

What changed from v1 (fit fix)

Item	v1	v2	Why
PCB→finger clear	-0.90 mm (overlap)	1.30 mm	Tabs can flex past board
Snap hinge	at parting rim	+2.0 mm into lid	Flex starts above headers
XY pocket clear	0.40	0.80	Board drops easily, still locates
Inner / outer W	26.2 / 30.2	30.4 / 34.4	Room for flex channel
Hook L / barb	6.2 / 0.90	7.0 / 0.95	Align with taller window
Base window	8.8×3.6	9.6×4.2	Lock with extra travel
PCB locate	posts only	posts + corner stops	Wide cavity still registers board
USB / LED / pins	USB-C, LED mem, enclosed	unchanged	Locked openings kept

Print orientation (no supports) K1C 0.20 / 0.4 nozzle

PETG preferred for snaps. 4-5 perimeters, 20% infill, no supports.



STL bbox bottom [54.1, 34.4, 13.4] lid [54.1, 34.4, 13.8] both << 215 mm

- BOM (no fasteners)
- 1 bottom.stl PETG/PLA 54.1×34.4×13.4 mm
 - 1 lid.stl PETG/PLA 54.1×34.4×13.8 mm
 - 0 screws snap-fit only
 - NodeMCU ESP-32S V1.1 + USB-C cable (existing)

- Assembly (tabs clear board, then click)
1. Seat board in bottom on 4 posts; corner stops locate XY. USB at U-slot; cable can stay plugged in.
 2. Lower lid: snap fingers travel in the side channels, clearing PCB / headers / USB / BOOT / EN.
 3. Four clicks as barbs enter the base windows.
 4. Open: pinch long-side windows, lift lid.

- CALIPER FLAGS (same as v1 – envelope unchanged)
1. PCB L × W (design 48.5 × 25.4 +0.8 pocket)
 2. Pin stick-out (design 6.0)
 3. USB-C overmold W × H seated (design 12.4 × 7.5)